

The logarithmic factor in the large-block Gaussian limit

Research proof candidate, pending independent review

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Let $R_\sigma^{(n,m)} = n!P(\sigma)$ be the likelihood ratio of a target order for m independent uniformly relabelled palindrome blocks. This note treats an iterated limit: first $m \rightarrow \infty$ for each fixed n , then $n \rightarrow \infty$. It makes no claim that the Gaussian approximation is uniform at the threshold $m \asymp n^{13/10}(\log n)^{1/5}$.

Theorem 1. *For every fixed $n \geq 3$, the finite vector*

$$(m^{5/2}(R_\sigma^{(n,m)} - 1) : \sigma \in S_n)$$

converges in distribution as $m \rightarrow \infty$ to a centered Gaussian vector $(G_\sigma : \sigma \in S_n)$. The expectations of the maximum absolute coordinates also converge. Moreover, for absolute constants $c, C > 0$,

$$cn^{13/4}\sqrt{\log n} \leq \lim_{m \rightarrow \infty} m^{5/2} \mathbb{E} \max_{\sigma \in S_n} |R_\sigma^{(n,m)} - 1| \leq Cn^{13/4}\sqrt{\log n}$$

for sufficiently large n .

1 The Gaussian limit

Put $d = n - 3$, $B_n = \binom{n}{3}$, and

$$w_a(p) = \binom{d}{a} p^a (1-p)^{d-a}, \quad \delta(l, x, r) = \frac{1}{2} - \frac{3}{2} \mathbf{1}\{x < \min(l, r)\}.$$

Let $Y = (Y_1, \dots, Y_n)$ consist of independent uniform $[0, 1]$ variables, and define

$$f_\sigma(p, Y) = \sum_{a=0}^d w_a(p) \delta(Y_{\sigma(a)}, Y_{\sigma(a+1)}, Y_{\sigma(a+2)}),$$

where target positions are numbered from zero. Each f_σ is centered as a function of Y . The triple projection is exactly

$$S_\sigma = B_n m^{-3} (1 - 1/m)^d \sum_{i=1}^m f_\sigma((i-1)/(m-1), Y_i),$$

with independent Y_i . Its scaled summands satisfy the Lindeberg condition for fixed n , since they are uniformly $O_n(m^{-1/2})$. Their covariance sums converge by ordinary Riemann integration. Therefore the multivariate Lindeberg central limit theorem gives the centered Gaussian limit with covariance

$$\mathbb{E} G_\sigma G_\tau = B_n^2 \int_0^1 \mathbb{E}[f_\sigma(p, Y) f_\tau(p, Y)] dp. \quad (1)$$

For clarity, the other terms vanish after scaling. At Hoeffding level ℓ , the blockwise centered kernel on any specified set of ℓ blocks is $O_n(m^{-3\ell})$. Symmetrization, or directly orthogonality of distinct canonical kernels for L^2 , bounds the full level by $O_n(m^{-5\ell/2})$ in L^2 . Levels $\ell \geq 2$ are therefore negligible. In level one the segments of length at least four give $O_n(m^{-7/2})$ in L^2 . Thus

$$m^{5/2} (R_\sigma^{(n,m)} - 1 - S_\sigma) \longrightarrow 0 \quad \text{in } L^2.$$

There are finitely many targets for fixed n , and their scaled coordinates have bounded second moments. Their maximum is consequently uniformly integrable, proving convergence of the expected maximum absolute value.

Equivalently, let W be an isonormal Gaussian process over the Hilbert space $L^2([0, 1] \times [0, 1]^n, dp dY)$. Then

$$G_\sigma = B_n W(f_\sigma).$$

Only a finite-dimensional subspace is needed here, so this is simply a convenient representation of a finite Gaussian vector with covariance (??).

2 The upper bound

The centered valley sequence in a fixed target has covariances $1/2, -1/4, 1/20$ at distances zero, one, and two, and zero beyond two. By Cauchy–Schwarz applied to shifted weight sequences,

$$\text{Var} \left(\sum_a w_a(p) \delta_a \right) \leq \frac{11}{10} \sum_a w_a(p)^2.$$

The exact beta-integral identity

$$\int_0^1 \sum_a w_a(p)^2 dp = \frac{4^d}{(2d+1)\binom{2d}{d}} \asymp n^{-1/2}$$

therefore gives $\text{Var } G_\sigma \leq Cn^{11/2}$. The elementary Gaussian exponential-moment bound for a maximum over $n!$ coordinates now implies

$$\mathbb{E} \max_\sigma |G_\sigma| \leq Cn^{11/4} \sqrt{\log(2n!)} \leq Cn^{13/4} \sqrt{\log n}.$$

No independence among these coordinates is required.

3 A canonical three-variable component

Let L, X, R be independent uniform variables, and let $u(L, X, R)$ be the component of $\delta(L, X, R)$ centered in each of its three variables:

$$u = \delta - \mathbb{E}(\delta \mid L, X) - \mathbb{E}(\delta \mid X, R) - \mathbb{E}(\delta \mid L, R) + \mathbb{E}(\delta \mid L) + \mathbb{E}(\delta \mid X) + \mathbb{E}(\delta \mid R).$$

It has zero conditional expectation given any two variables. Its variance is

$$\mathbb{E}u^2 = \frac{1}{20}. \tag{2}$$

Here is an explicit check. With $g(t) = \frac{3}{4}(1-t)^2 - \frac{1}{4}$, the one-variable projections are $g(L), -2g(X), g(R)$ and $\mathbb{E}g^2 = 1/20$. The variances of the two-variable conditional expectations are

$$\text{Var } \mathbb{E}(\delta \mid L, R) = \frac{1}{8}, \quad \text{Var } \mathbb{E}(\delta \mid L, X) = \text{Var } \mathbb{E}(\delta \mid X, R) = \frac{5}{16}.$$

Orthogonality of the Hoeffding components gives $\mathbb{E}u^2 = \frac{1}{2} + 6/20 - \frac{1}{8} - 2(5/16) = 1/20$.

4 An independent Gaussian assignment matrix inside the process

Choose $N \asymp n$ movable positions in the middle third of a base target, spaced by at least three positions. Keep all other labels fixed as anchors, and allow the N movable labels to permute arbitrarily among the movable positions. Let $\mathcal{F}$ be this family of $N!$ target orders. Each movable position v has two

fixed adjacent anchor labels λ_v, ρ_v . These anchor pairs are disjoint between positions, and no anchor is a movable label.

For a movable label b , define the Gaussian entry

$$Z_{v,b} = B_n W(w_{v-1}(p)u(Y_{\lambda_v}, Y_b, Y_{\rho_v})).$$

These N^2 entries are mutually independent. Indeed, distinct entries use different triples of underlying variables, with at most two variables in common. Integrating in a variable present in only one triple makes their inner product zero by the canonical property of u . Joint Gaussianity turns this orthogonality into independence. For each row v , all entries have the same variance, and

$$\text{Var } Z_{v,b} = \frac{B_n^2}{20} \int_0^1 w_{v-1}(p)^2 dp \asymp n^{9/2}.$$

The constants are uniform over the chosen central positions: the integral is $\binom{d}{v-1}^2 / ((2d+1)\binom{2d}{2v-2}) \asymp n^{-3/2}$.

If a target $\sigma \in \mathcal{F}$ assigns label $\pi(v)$ to position v , its orthogonal projection onto the span of these Gaussian entries is precisely $\sum_v Z_{v,\pi(v)}$. To check this, a target triple can have a nonzero inner product with the defining canonical kernel only if it contains all three labels λ_v, b, ρ_v . The two fixed anchors are separated by position v , so the only such consecutive triple has center label b assigned to v . Its coefficient is exactly w_{v-1} . Consequently,

$$G_\sigma = \sum_v Z_{v,\pi(v)} + H_\sigma,$$

where the entire centered Gaussian residual vector (H_σ) is independent of the matrix Z .

Conditional Jensen for the maximum gives

$$\mathbb{E} \max_{\sigma \in \mathcal{F}} G_\sigma \geq \mathbb{E} \max_{\pi \in S_N} \sum_v Z_{v,\pi(v)}.$$

A greedy assignment proves a lower bound for the right side: process rows one by one, and select in each row the largest entry among the remaining columns. The remaining set depends only on previous rows, so it is independent of the current row. For the first $N/2$ rows, at least $N/2$ independent centered Gaussians of common standard deviation $\asymp n^{9/4}$ remain. Their expected maximum is at least $cn^{9/4}\sqrt{\log N}$. Later row increments have nonnegative expectation. Thus

$$\mathbb{E} \max_{\sigma \in S_n} |G_\sigma| \geq cNn^{9/4}\sqrt{\log N} \geq cn^{13/4}\sqrt{\log n}.$$

Together with the Gaussian upper bound and the convergence of expectations, this proves Theorem ??.

Interpretation. The logarithmic factor in the uniform upper estimate reflects real behavior of this random construction in its Gaussian limit. Extending this lower bound uniformly down to $m \asymp n^{13/10}(\log n)^{1/5}$ would require additional quantitative comparison or direct assignment estimates. The separate adjacent-swap argument already proves a uniform lower bound without the logarithm in the regime $m \geq Cn^{13/10}$.